INCH-POUND

MIL-PRF-1/299H
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SUPERSEDING
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PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING
TYPE 5927

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: TR, bandpass, frequency range 3,100 to 3,500 MHz, incident power 825 kw.

ABSOLUTE RATINGS:

| Parameter: | Incident power | Ii | Ebb | Du | Alt |
|---|---|---|---|---|---|
| Unit: | kw | μA dc | V dc | - - - | Ft |
| Maximum: | - - - | 200 | -700 | 0.001 | 10,000 |
| Minimum: | 100 | - - - | -500 | - - - | - - - |

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1.
Mounting: 153-JAN (or equivalent) 268-JAN.
Mounting position: Any.

TEST CONDITIONS:

| Parameter: | Incident power | Ii | tp1 | tp2 | F | Du |
|---|---|---|---|---|---|---|
| Unit: | kw | μA dc | μs | μs | - - - | - - - |
| Test condition 1: | | | | | | |
| Maximum: | - - - | - - - | 1.15 | - - - | - - - | - - - |
| | 50 | - - - | 1.00 | - - - | F2 | 0.001 |
| Minimum: | - - - | - - - | 0.85 | - - - | - - - | - - - |
| Test condition 2: | | | | | | |
| Maximum: | 220 | - - - | 1.15 | 0.60 | - - - | - - - |
| | 200 | 200 | 1.00 | 0.50 | F2 | 0.001 |
| Minimum: | 180 | - - - | 0.85 | 0.40 | - - - | - - - |
| Test condition 3: | | | | | | |
| Maximum: | 825 | - - - | - - - | - - - | - - - | - - - |
| | 750 | 200 | - - - | - - - | F3 | 0.001 |
| Minimum: | 675 | - - - | - - - | - - - | - - - | - - - |

| Frequency | | |
|---|---|---|
| F | MHz | ± |
| F1 | 3,100 | 0.1 % |
| F2 | 3,300 | 0.1 % |
| F3 | 3,425 | 0.1 % |
| F4 | 3,500 | 0.1 % |

GENERAL:
Qualification - Required.

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

FSC 5960

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TABLE I. Testing and inspection.

| Inspection | MIL-STD-1311 Method | Notes | Test | Conditions | Symbol | Limits | | Units |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Min | Max | |
| Qualification inspection | | | | | | | | |
| Degradation due to vibration | 4021 | - - - | - - - | | - - - | - - - | - - - | - - - |
| Low-level VSWR | 4473 | 2, 3 | - - - | σ' = 1.05 | σ | - - - | 1.9 | - - - |
| High-level VSWR | 4474 | 4, 5 | 1 | | σ | - - - | 1.15 | - - - |
| Humidity | 1011 | - - - | - - - | | - - - | - - - | - - - | - - - |
| Conformance inspection, part 1 | | 1 | - - - | | | | | |
| Ignitor voltage drop | 4401 | 6 | - - - | | t | - - - | 5.0 | sec |
| Spike leakage energy | 4406 | 6 | - - - | Ii = 200 μA dc | Eid | -250 | -400 | V dc |
| Flat leakage power | 4452 | 4 | 2 | | Ws | - - - | 0.3 | erg |
| | 4452 | 3 | 2 | | pf | - - - | 50 | mw |
| Low-level VSWR | 4473 | 3 | - - - | σ' = 1.05 (max)<br>F1 and F4<br>F2 | <br>σ<br>σ | <br>- - -<br>- - - | <br>1.60<br>1.40 | <br>- - -<br>- - - |
| Conformance inspection, part 2 | | | | | | | | |
| Dielectric material strain | 4401 | - - - | - - - | | - - - | - - - | - - - | - - - |
| Insertion loss (fixed tuned) | 4416 | 3 | - - - | F = F2;<br>Ii = 0 | Li | - - - | 0.7 | dB |
| Ignitor interaction (insertion loss) | 4421 | 6 | - - - | F = F2;<br>Ii = 200 μA dc | ΔLi | - - - | 0.3 | dB |
| Recovery time | 4471 | 4, 8 | 2 | | t | - - - | 10 | μs |
| Bump | 1036 | 7 | - - - | Hammer angle, 25° | - - - | - - - | - - - | - - - |
| Conformance inspection, part 3 | | | | | | | | |
| Life test | - - - | 4 | 3 | Group D;<br>Ii = 150 μA dc | t | 500 | - - - | hrs |
| Life-test end points: | - - - | - - - | - - - | | | | | |
| Recovery time | 4471 | 4, 8 | 3 | | t | - - - | 30 | μs |
| Flat leakage power | 4452 | 3 | 2 | | pf | - - - | 50 | mW |
| Spike leakage energy | 4452 | 4 | 2 | | Ws | - - - | 0.3 | erg |
| Insertion loss (fixed tuned) | 4416 | 3 | - - - | F = F2; Ii = 0 | Li | - - - | 1.2 | dB |

See notes at end of Table I.



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NOTES:

1. This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

2. The input standing-wave ratio shall be measured at intervals of 30 MHz from F1 - 30 MHz to F4 + 30 MHz. At no frequency in this interval shall the voltage standing wave ratio be greater than the amount specified.

3. This test shall be performed using the flanges specified in Drawing 268-JAN.

4. This test shall be performed using the mount specified in Drawing 153-JAN (or equivalent).

5. With a load standing wave ratio of less than 1.05 the voltage standing ratio produced by the tube shall be less than the amount specified.

6. The ignitor power supply shall have an open circuit voltage of 600 V dc negative with respect to the tube body. The ignitor series resistance shall be 1.60 MegΩ. The ignitor shall fire within the specified time after application of voltage.

7. Anvil in accordance with Drawing 188-JAN. An equivalent test employing MIL-STD-202-213 may be used provided G level and pulse duration of shock pulse level are equivalent to that obtained using MIL-STD-1311, method 1036 with hammer angle of 25°

8. The recovery time shall be measured after 30 minutes of operation under the conditions specified.



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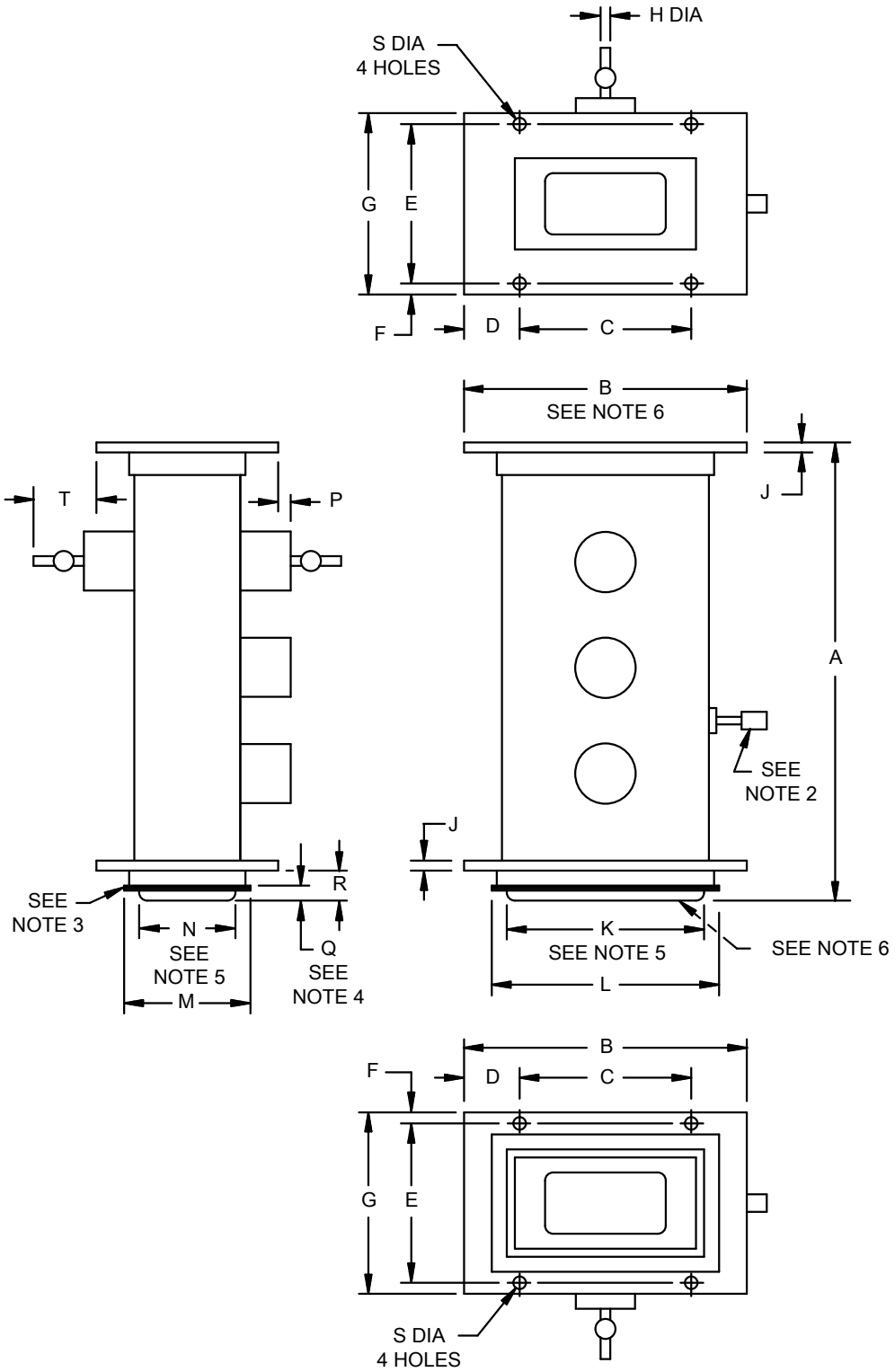



FIGURE 1. Outline drawing of electron tube type 5927.

See notes at end of Figure 1.



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| Dimensions | | | | |
|---|---|---|---|---|
| Ltr | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| | Conformance inspection, part 1 | | | (see note 1) |
| A | 4.828 | 4.888 | 122.63 | 124.16 |
| K | | 3.008 | | 76.40 |
| P | | .187 | | 4.76 |
| Q | .165 | .195 | 4.19 | 4.95 |
| R | .347 | .377 | 8.81 | 9.58 |
| T | | .812 | | 20.64 |
| | Conformance inspection, part 2 | | | |
| C | 2.490 | 2.510 | 63.25 | 63.75 |
| E | 2.240 | 2.260 | 56.90 | 57.40 |
| L | 3.203 | 3.233 | 81.36 | 81.85 |
| M | 1.703 | 1.733 | 43.26 | 44.02 |
| N | | 1.508 | | 38.30 |
| | Reference dimensions | | | |
| B | 4.125 | | 104.78 | |
| D | .812 | | 20.64 | |
| F | .187 | | 4.76 | |
| G | 2.625 | | 66.68 | |
| H | .250 | | 6.35 | |
| J | .125 | | 3.18 | |
| S | .218 | | 5.56 | |

NOTES:

1. This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.
2. Exhaust tube shall not extend beyond flange more than 0.25 (6.35 mm) inch.
3. Gasket in accordance with Drawing 189 JAN. Gasket to be securely attached.
4. Dimension Q shall be measured prior to the attachment of the gasket to the tube.
5. Edges may be rounded.
6. Nickel plating or Rhodium flash over silver plating optional. If nickel plating is required, it is recommended that it be used only when other platings cannot meet performance requirements. Surface plating shall meet and enable the Gas Switching tube to meet all interface and performance requirements.

FIGURE 1. Outline drawing of electron tube type 5927 - Continued.



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Referenced documents. In addition to MIL-PRF-1, this document references the following:

MIL-STD-202-213
MIL-STD-1311
Drawing 153-JAN
Drawing 188-JAN
Drawing 189 JAN
Drawing 268-JAN

NOTE: To obtain copies of JAN drawings, please send a request via email to TubesAmps@dla.mil.

Amendment notations. The margins of this specification sheet are marked with vertical line to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2025-010)

Reviewing activities:
Army - AR
Navy - AS, CG, MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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